



AI Voice Detection & Cryptographic Evidence Infrastructure

Complete Technical Architecture, DSP Feature Extraction & Decision Science

Executive Summary: VOXTRACE is an enterprise-grade cybersecurity and digital authenticity platform designed to protect financial institutions, executive communication, call centers, and media platforms against real-time AI voice cloning and deepfake fraud. This document provides a complete mathematical, architectural, and operational explanation of how VOXTRACE captures, cleans, analyzes, ranks, and immutably anchors voice recordings.

1. System Overview & Threat Model

Modern generative speech synthesis (such as ElevenLabs, VALL-E, OpenVoice, Bark, and Real-time Voice Conversion / RVC) utilizes neural vocoders and diffusion models to reconstruct human speech from reference audio clips as short as 3 seconds. Attackers weaponize these models to execute **CEO Fraud / Business Email Compromise (BEC)**, family emergency impersonations, KYC/biometric bypass, and disinformation campaigns.

Traditional audio filters and generic machine learning models suffer from severe false-positive rates when encountering background noise, compressed cellular codecs (e.g., AMR, AWB), singing, or muffled speech. VOXTRACE solves this through a deterministic **5-Stage Acoustic & Digital Signal Processing (DSP) Pipeline** paired with an **8-Factor Biological Voice Marker Consensus Engine**.

The 4 Core Pillars of VOXTRACE:

- **Detect:** Multi-stage audio feature extraction evaluating biological vocal tract dynamics and vocoder artifacts.
- **Explain:** Every decision is broken down into sub-indicators (Jitter, Shimmer, Intonation, Breath, Spectrum) so analysts know *why* a sample is flagged.
- **Verify:** Deterministic scoring calibrated across speech, song, and compressed codecs with instant verification.
- **Preserve:** Generates a cryptographic SHA-256 Evidence Passport anchored in a tamper-evident audit log.

2. Universal Audio Intake & Transcoding

Voice traffic originates from diverse environments: phone calls (AMR-NB / AMR-WB), WhatsApp voice notes (Opus), call center recordings (WAV / MP3), and browser mics (WebM). Standard browser Web Audio API fails when decoding cellular formats like `.amr` or `.awb`.

Audio Codec / Ext	Sample Rate	Source Environment	Intake & Transcoding Path
.WAV / .MP3 / .M4A	16 kHz – 48 kHz	Studio / PC / Call Center	Native Web Audio <code>AudioContext.decodeAudioData</code> (0ms overhead)
.AMR (AMR-NB)	8,000 Hz	2G/3G Cellular & WhatsApp	Auto-transcoded via <code>/api/audio/convert</code> (FFmpeg PCM 16kHz)
.AWB (AMR-WB)	16,000 Hz	VoLTE HD Voice Phone Calls	Auto-transcoded via <code>/api/audio/convert</code> (FFmpeg PCM 16kHz)
.3GP / .3GPP	8 – 16 kHz	Mobile Call Recorders	Auto-transcoded via <code>/api/audio/convert</code> (FFmpeg PCM 16kHz)
WebRTC / Mic Live	44.1 / 48 kHz	Live In-Browser Verification	Captured via <code>navigator.mediaDevices.getUserMedia</code> + Canvas Visualizer

3. The 5-Stage Analysis Pipeline

When an audio stream or file enters the engine, it undergoes sequential filtration, decomposition, and contextual evaluation. No single heuristic makes the final determination; instead, decisions emerge from multi-dimensional consensus.

0

Stage 0: Silence & Energy Guard (Anti-False-Positive Filter)

Computes Root-Mean-Square (RMS) energy and Peak Amplitude across all PCM samples. If `RMS < 0.008` or `Peak < 0.025`, the audio is rejected immediately with a score of **4/100** and verdict **"NO SPEECH DETECTED"**. This prevents ambient noise or muted microphones from receiving artificial high scores.

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Stage 1: Frame-Level DSP & Spectral Feature Extraction

Segments audio into 2048-sample frames with a 512-sample hop (75% overlap) shaped by a **Hamming Window**: `w[i] = 0.54 - 0.46*cos(2πi / (FL-1))`. Computes Fast Fourier Transform (FFT), sub-band energies (Low: <500Hz, Mid: 500-3000Hz, High: >3000Hz), Spectral Centroid, Spectral Flatness, and Zero Crossing Rate (ZCR).

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Stage 2: Audio Context & Type Classification

Examines pitch span, voiced ratio, and centroid spread to determine the operational context:

- **SPEECH**: Conversational human speech (evaluates prosody and speech naturalness).
- **SONG**: Singing vocals with wide melodic ranges (shifts to singing-specific weights).
- **MUSIC**: Pure instrumental content (returns neutral 50/100 without false AI accusation).
- **LOW_SPEECH**: Whispers or clicks without sustained voice (flags "INSUFFICIENT SPEECH").

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Stage 3: 8-Factor Deep Acoustic Extraction

Evaluates 8 distinct biological vocal tract markers against synthetic neural vocoder signatures. Cycle-to-cycle pitch periods, amplitude modulation, phoneme evolution, and pulmonary breath transitions are quantified into calibrated sub-scores (0 to 100).

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Stage 4: Consensus Scoring & Tamper-Proof Anchoring

Aggregates weighted indicator sub-scores. Applies **AI Consensus Penalties** (if ≥4 indicators flag synthetic patterns, score is hard-capped <35) or **Human Floor Bonuses** (if ≥5 markers demonstrate biological complexity, score is protected ≥74). Anchors results with SHA-256 hashes into the immutable ledger.

4. Mathematical Foundations of Audio Extraction

A. Normalized Autocorrelation Pitch Extraction (F0)

Calculated over lags corresponding to 70 Hz to 800 Hz to cover both deep bass male speech and high soprano singing:

$$r(\text{lag}) = \sum [x(i) \cdot x(i+\text{lag})] / \text{sqrt}(\sum [x(i)]^2 \cdot \sum [x(i+\text{lag})]^2)$$

B. Spectral Flatness (Wiener Entropy)

Measures whether energy is concentrated in tonal harmonics (human) or white/diffused vocoder noise (synthetic):

$$\text{Flatness} = \exp\left(\frac{1}{N} \cdot \sum \ln(m_k)\right) / \left(\frac{1}{N} \cdot \sum m_k\right)$$

C. Spectral Centroid (Timbral Brightness)

Center of mass of the frequency spectrum:

$$\text{Centroid} = \left(\sum [k \cdot m_k] / \sum m_k\right) \cdot (\text{SampleRate} / \text{FrameLength})$$

5. The 8 Biological Voice Markers (AI vs Human)

Human vocal production requires coordinated interaction between the lungs (airflow), vocal folds (laryngeal vibration), and vocal tract articulators (tongue, lips, velum). Neural vocoders generate audio by predicting waveform chunks or spectrogram inversions, leaving distinct mathematical artifacts.

Marker	Biological Mechanism (Human )	Synthetic Signature (AI Clone )
1. Pitch Jitter (Cycle-to-Cycle F0)	Vocal cords cannot sustain mathematically invariant vibration. Micro-muscle tremors cause natural cycle variation of 0.4% to 2.5% .	AI vocoders produce unnaturally smooth, stable pitch tracks with jitter < 0.2% (robotic consistency).
2. Amplitude Shimmer	Subglottal pressure naturally fluctuates between vocal fold opening cycles, producing 3% to 12% shimmer.	Diffusion/autoregressive speech outputs have uniform energy profiles with shimmer < 1% .
3. Spectral Dynamics & Evolution	Every phoneme transition reconfigures oral cavity shape, generating continuous spectral centroid and flux variation.	Spectrograms exhibit static formant lines and repetitive spectral flatness typical of vocoder frame concatenation.
4. Prosodic Intonation Arc	Speech carries emotional, semantic, and question/statement pitch contours (pitch CV typically 7% to 20%).	Either monotone flat delivery (pitch CV < 3%) or exaggerated, repetitive sinusoidal intonation arcs.
5. Breath & Pause Acoustics	Humans must inhale. Shabdon ke beech inhalation noise (15-25 dB lower than voice) and room reflections exist.	Digital silence or abrupt gate cuts between words; zero inhalation sounds detected.
6. High-Frequency Continuity	Fricatives ( ,  , ) create turbulent air noise extending smoothly into the 8 kHz – 16 kHz band.	Abrupt frequency roll-off above 4kHz/8kHz due to neural vocoder downsampling or synthetic metallic high-frequency haze.
7. Temporal Energy Dynamics	Natural speech alternates strongly between stressed (accented) and unstressed syllables across sentences.	Metronome-like regularity or flat energy envelopes typical of sentence-level neural text-to-speech.
8. Voiced / Unvoiced Transitions	Zero Crossing Rate (ZCR) shifts rapidly as speech moves from periodic vowels (low ZCR) to aperiodic consonants (high ZCR).	ZCR remains suspiciously uniform across transitions, failing to model turbulent oral airflow correctly.

Weight Allocation in Final Trust Score Calculation

Speech Mode Weights:

- **Pitch Jitter:** 18%
- **Breath & Pauses:** 16%
- **Spectral Evolution:** 16%
- **Amplitude Shimmer:** 14%
- **Prosodic Intonation:** 12%
- **High-Frequency Continuity:** 10%
- **Temporal Dynamics:** 10%
- **ZCR Transitions:** 4%

Song / Singing Mode Weights:

- **Vocal Pitch Jitter:** 18%
- **Spectral Richness:** 18%
- **Melodic Intonation Arc:** 16%
- **Amplitude Shimmer:** 15%
- **Breath Between Phrases:** 14%
- **High-Frequency Presence:** 10%
- **Vocal Dynamics:** 9%

6. Multi-Factor Consensus & Verdict Classification

A simple weighted average is insufficient for forensic cybersecurity because an advanced clone might fool 2 markers while failing miserably on 3 others. VOXTRACE implements a **Non-Linear Consensus Engine**:

AI Consensus Rule (Hard Penalty Capping):

- If **5 or more** AI signals trigger: Trust Score is hard-capped at ≤ 22 (**LIKELY AI-GENERATED**).
- If **4 AI signals** trigger: Trust Score is hard-capped at ≤ 35 .
- If **3 AI signals** trigger: Trust Score is capped at ≤ 50 (**SUSPICIOUS**).

Human Consensus Rule (False-Positive Protection):

- If **5 or more** biological markers show strong human complexity (natural jitter, healthy shimmer, breath transitions), the Trust Score is protected with a floor of ≥ 74 (**LIKELY GENUINE**), even if minor background noise distorted one metric.

Trust Score	Verdict	Risk Level	Operational Action
75 – 99	LIKELY GENUINE	Low Risk	Approved for execution, verification passport issued.
55 – 74	PROBABLY GENUINE — MONITOR	Moderate	Acceptable for standard operations; flag for continuous monitoring.
40 – 54	SUSPICIOUS — REVIEW REQUIRED	High	Hold transaction/action. Secondary verification step required.
02 – 39	LIKELY AI-GENERATED	Critical	Immediate block. Voice clone attack logged; forensic evidence preserved.
Special	NO SPEECH / SILENCE	Invalid	Audio energy below threshold. User prompted to speak clearly.

7. Cryptographic Evidence Passport & Blockchain Proof

Detecting a deepfake is only half the battle; organizations must present legally defensible, tamper-evident proof to compliance officers, law enforcement, and insurance underwriters. VOXTRACE issues an **Evidence Passport** for every verification.

Evidence Passport Structure:

- Evidence ID:** Deterministic identifier (e.g. **VT-260923-9F4B**) derived from date and hash.

- **SHA-256 Content Hash:** Exact cryptographic digest of the raw audio bytes (verified byte-for-byte).
- **Block & Merkle Link:** Linked to previous verification block using cryptographic chaining (`prevHash + evidenceHash + timestamp`).
- **Tamper-Proof Verification:** If anyone modifies a single millisecond of the audio recording later, re-computing the SHA-256 fails instantly against the Evidence Passport.

8. Production Integration & API Usage

VOXTRACE integrates seamlessly into enterprise voice pipelines:

```
curl -X POST https://voxtrace.vercel.app/api/verify \ -H "Authorization: Bearer vt_live_sec_key" \ -F "audio=@call_recording.awb"
```

Returns JSON response containing `trustScore`, `verdict`, `audioType`, `indicators` breakdown, and `evidencePassport`.